

Cosmic Structural Tension II

A dimensionless boundary index for gravitational-basin membership in Λ CDM

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Abstract

We propose Cosmic Structural Tension II (TEC-2), a phenomenological Λ CDM-compatible framework in which dark energy does not destroy bound structures internally but defines an outer boundary of gravitational-basin membership. The central diagnostic is a dimensionless local-global index comparing the effective acceleration associated with the cosmological constant to the mean gravitational acceleration of an enclosed mass.

$$T\Lambda(R,M) = \Omega\Lambda H_0^2 R^3 / (GM)$$

We test the index in a pilot literature sample of 11 published structures, including Local Group, M81/M82, Fornax-Eridanus, Virgo, Coma, and several collapsing supercluster cores. Among 9 turnaround-like or zero-velocity entries, all central estimates satisfy $T\Lambda \leq 1$, with median $T\Lambda = 0.319$. This does not prove universality, but provides preliminary independent consistency with TEC-2 as an exterior-boundary criterion.

1. Physical Idea

The guiding claim is deliberately limited: TEC-2 does not introduce a new force. It reorganizes a known Λ CDM boundary scale into an operational index. The phrase 'structural tension' refers to the balance between local gravitational aggregation and global accelerated expansion. In this view, galaxies, halos, clusters and superclusters are nested gravitational basins; black holes are extreme inner basins, not exceptions to the framework.

The proposed publication-level statement is: dark energy does not destroy gravitationally bound structures; it defines the outer limit of gravitational membership. The testable question is whether observed boundaries such as zero-velocity or turnaround surfaces lie at or below $T\Lambda \approx 1$.

2. Formal Derivation Of The Base Index

The base index follows from comparing two accelerations. For a mass M enclosed within physical radius R , the characteristic inward gravitational acceleration is

$$a_g(R,M) = GM/R^2$$

For a cosmological constant in the low-redshift Λ CDM limit, the outward acceleration scale associated with dark energy is

$$a_\Lambda(R) \approx \Omega\Lambda H_0^2 R$$

The boundary regime is obtained when a_Λ and a_g are comparable. Their ratio defines a dimensionless continuous diagnostic:

$$T\Lambda(R,M) = a_\Lambda/a_g = \Omega\Lambda H_0^2 R^3/(GM)$$

Thus $T\Lambda < 1$ indicates that local gravity dominates the analysed scale; $T\Lambda \approx 1$ marks the exterior boundary regime; $T\Lambda > 1$ indicates that the analysed scale lies beyond the gravitational basin in the ideal isolated limit.

3. Relation To Existing Λ CDM Boundaries

For a cosmological constant, $T\Lambda = 1$ is equivalent to the maximum turnaround-radius scale in Schwarzschild-de Sitter/Kottler and spherical-collapse arguments:

$$R_{ta,max} = (3GM/\Lambda c^2)^{1/3} = [GM/(\Omega\Lambda H_0^2)]^{1/3}$$

Thus TEC-2 should not be presented as a replacement for turnaround theory, virial radii, spherical collapse or Λ CDM. Its proposed contribution is classification: a continuous dimensionless index applicable across inner cores, virialized regions, turnaround-like surfaces, and large-scale structures.

Known framework	What is already known	TEC-2 contribution
Λ CDM turnaround	There is a maximum mass-dependent exterior scale for bound structures.	Recasts the scale as a continuous membership index $T\Lambda$.
Spherical collapse	Ideal overdensities pass through expansion,	Uses the boundary idea as a classifier

Known framework	What is already known	TEC-2 contribution
	turnaround, collapse and virialization.	across object classes.
Virial radius	Describes an internal approximately relaxed halo region.	Separates internal relaxation from exterior gravitational-basin membership.
Cosmic-web classifiers	Classify knots, filaments, sheets and voids by density/tidal geometry.	Adds an acceleration-ratio criterion tied to dark-energy dominance.

4. Extensions: Dynamic Dark Energy And Environment

The natural extension is not to invent a new force, but to generalize the acceleration ratio. For a dark-energy equation of state $w(z)$, the repulsive contribution in the acceleration equation is proportional to $-[1 + 3w(z)]/2$. A dynamical dark-energy version of the index is therefore

$$TDE(R,M,z) = \{-[1 + 3w(z)]/2\} \Omega_{DE}(z) H(z)^2 R^3/(GM)$$

For $w = -1$ and $z = 0$, this reduces to TA . This version asks a sharper question: if dark energy evolves with time, do the boundaries of gravitational basins evolve measurably with redshift?

A later, more realistic version should include environmental corrections, because real structures are not isolated spheres. We propose the following future extension, not yet validated in this preprint:

$$T_{basin}(R,M,z,\eta) = TDE(R,M_{eff},z) \cdot F_{env}(\eta)$$

Here $M_{eff}(R)$ is the lensing or dynamical mass enclosed by the basin, and $F_{env}(\eta)$ is an environmental factor describing isolation, filamentarity, tidal field or proximity to neighbouring masses. This is the most original future direction, but it must be fitted from simulations or survey data before being treated as physical.

5. Pilot Data And Radius Definitions

The pilot sample is intentionally small and heterogeneous. Each entry is labelled by radius type, because mixing virial radius, zero-velocity radius, turnaround radius and dense collapsing core radius would otherwise overstate the evidence. Core or virial points are controls; turnaround-like and zero-velocity points are the informative boundary tests.

Object	Radius type	M [$M_{\odot}$]	R [Mpc]	TA	Uncertainty flag	Primary source
Milky Way / Leo I	satélite ligado, não turnaround	1.70e+12	0.25	0.007	no	Pavlidou & Tomaras 2014
Local Group	turnaround local	4.17e+12	0.96	0.154	no	Karachentsev & Kashibadze 2006
M81/M82 Group	turnaround local	1.60e+12	0.89	0.319	no	Karachentsev et al. 2002
Fornax-Eridanus Complex	turnaround aproximado	1.92e+14	4.60	0.367	no	Nasonova et al. 2011
Virgo Cluster	turnaround local	1.10e+15	8.60	0.418	no	Karachentsev & Nasonova 2010
Virgo Cluster X-ray core	raio interno, não turnaround	1.40e+14	1.20	0.009	no	Urban et al. 2011
Coma Cluster	turnaround/infall bound; ambiente filamentar	1.90e+15	6.66	0.112	possible	Benisty et al. 2025 preprint
A2142 supercluster core	turnaround/collapsing core	3.41e+15	11.87	0.355	no	Einasto et al. 2018/2021
Sloan Great Wall core	turnaround/collapsing core	2.67e+15	11.13	0.373	no	Einasto et al. 2016/2021
Shapley supercluster core	turnaround/collapsing core	1.93e+16	18.40	0.234	no	Reisenegger et al.

Object	Radius type	M [M_sun]	R [Mpc]	TΛ	Uncertainty flag	Primary source
Corona Borealis core	turnaround/candidate bound supercluster	4.45e+16	18.55	0.104	no	Pearson et al. 2014

6. Results

All 9 central turnaround-like estimates satisfy $T\Lambda \leq 1$. The largest central value is $T\Lambda = 0.418$. Coma is the only object whose uncertainty envelope can approach or exceed the boundary, consistent with its known filamentary, non-isolated environment and uncertainty in the outer infall/turnaround scale.

Validação piloto TEC-2 com estruturas publicadas

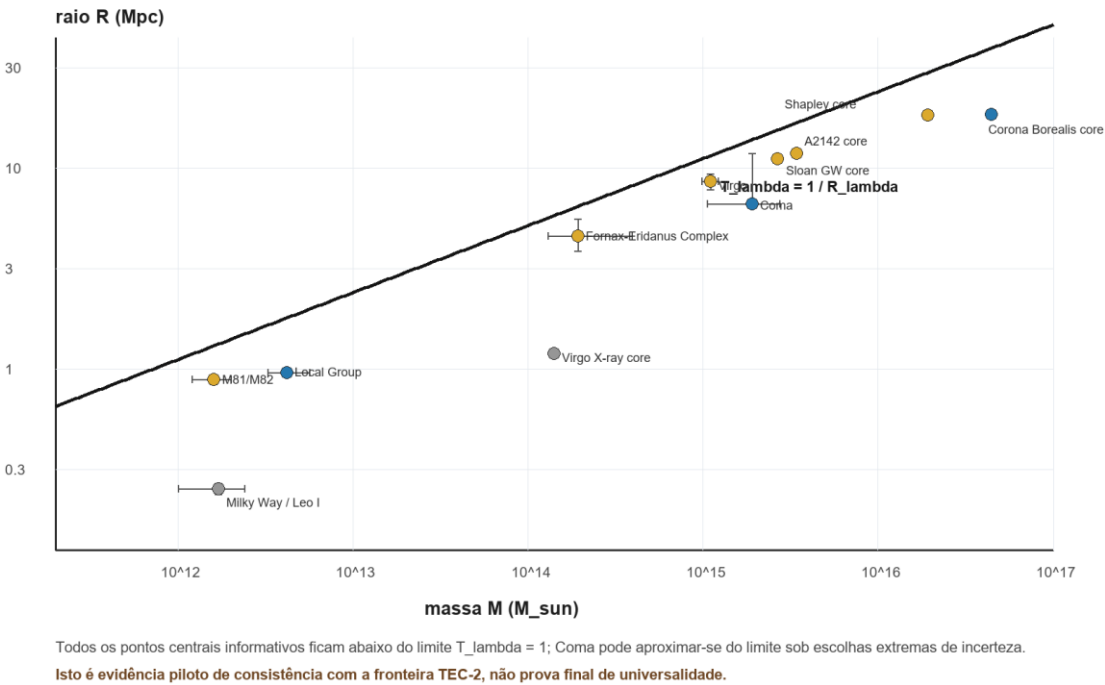


Figure 1. Published structures in the mass-radius plane. Error bars show reported or adopted mass/radius uncertainty ranges. The solid line is $T\Lambda = 1$.

Teste sintético: recuperando a escala $R \sim M^{1/3}$

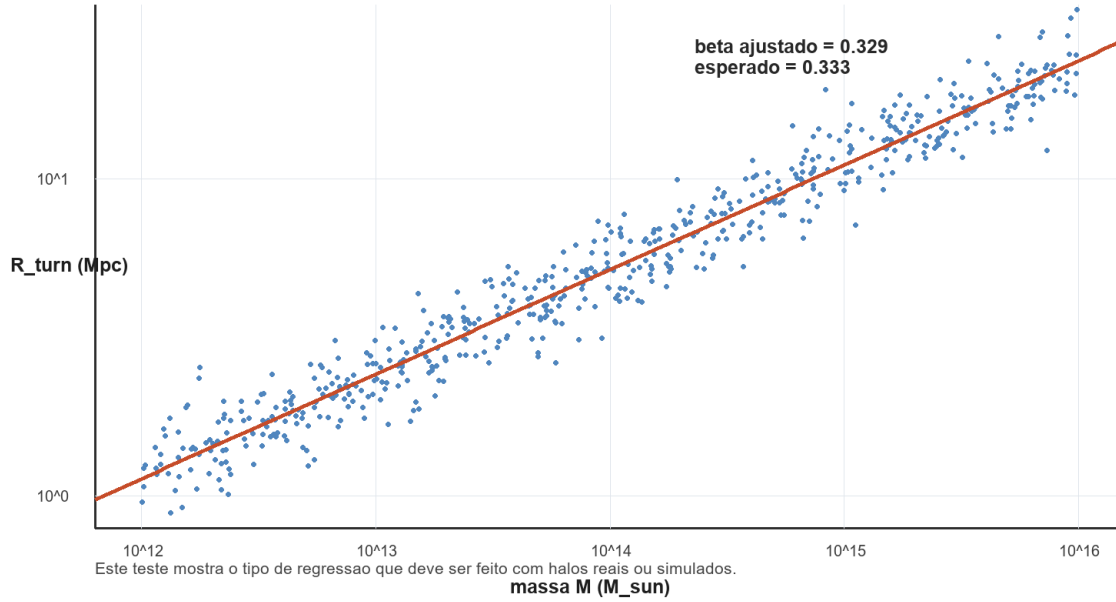


Figure 2. Synthetic sanity check recovering the expected $R \propto M^{1/3}$ scaling. This is a methodological check, not observational proof.

7. Discussion

The pilot result supports a cautious claim: structures already interpreted in the literature as bound, in turnaround, or in collapse are consistent with the TEC-2 boundary criterion. The result is not yet a discovery of new physics. It is a compact re-expression of a known Λ CDM scale that appears useful as a gravitational-basin classifier.

The next decisive test is comparative. TA must be evaluated against existing metrics: mass alone, mean density, virial radius, turnaround radius, phase-space caustics, and cosmic-web classifiers. A publishable follow-up should test whether TA improves classification or reduces scatter in boundary prediction using a larger homogeneous simulation or survey sample.

8. Limitations

The present sample is small, heterogeneous, and partly based on different definitions of boundary radius. Some masses and radii are model dependent, and supercluster cores are not spherical isolated systems. The current analysis therefore provides preliminary independent consistency, not universal proof.

9. Conclusion

The validation pilot shows that published structures interpreted as bound, in turnaround, or collapsing remain below $TA = 1$ in their central estimates, consistent with TEC-2 as an outer-boundary criterion for gravitational membership. This result does not demonstrate universality, but it gives the hypothesis a defensible, testable preprint-level form.

References

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